

When does DoRA help?

Geometry, budgets, and held-seed few-shot adaptation

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Why separate magnitude from direction?

LoRA adds a low-rank update BA , coupling row direction and row norm. DoRA writes $W = m \odot (W_0 + BA) / \|W_0 + BA\|_2$, so BA controls direction while m controls row scale.

Research question: does this extra degree of freedom improve target adaptation after fair selection, budget controls, and negative controls?

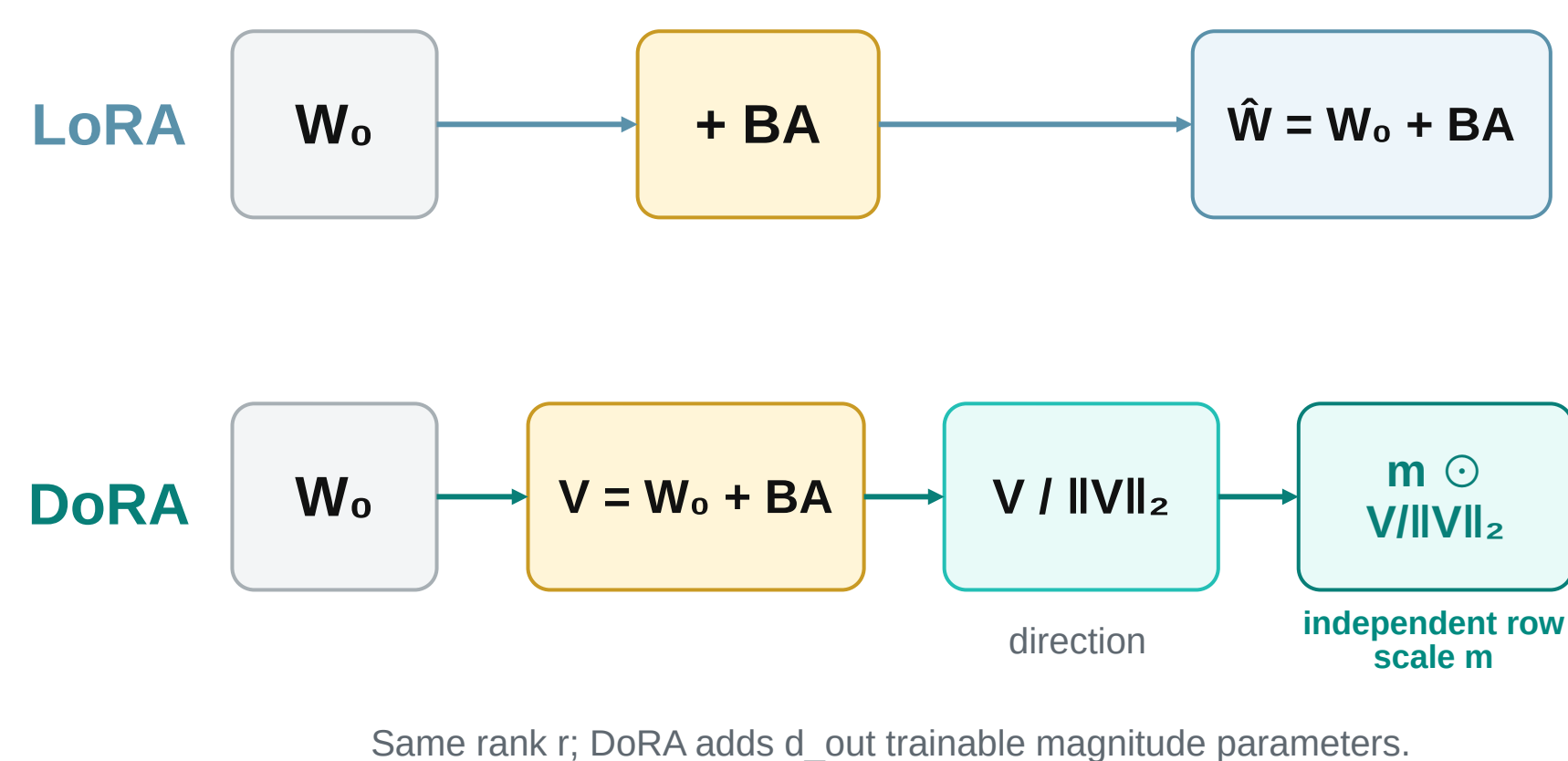


Figure 1. LoRA couples scale and direction; DoRA models row scale explicitly.

Frozen protocol before held-seed evaluation

Pilot only (validation): learning rate, early stopping, and rank allocation.

Held-seed: 20 MLP + 10 CNN adaptation seeds; one fixed checkpoint each; 3 shifts; Holm families $m=9 / 6 / 3$ (secondary descriptive).

Baselines: frozen, magnitude-only, LoRA, LoRA+ (fixed $B/A=16$), matched-budget LoRA, budgeted DoRA, and full fine-tuning.

Robustness: 4 nested data regimes; 10 synthetic targets × 5 initializations.

1,960 saved result rows; 1,840 training / optimization jobs.

Negative controls prevent a victory-only story

Contrast (MLP): magnitude-only 96.90% > DoRA 95.18%.

Rotation (MLP): matched-budget LoRA = DoRA = 93.89%.

Mixed (MLP): matching the parameter budget reduces the DoRA gap to +0.53 pp; its CI crosses zero.

Scope: Digits, one fixed pretrained base per architecture, and one target split already seen during exploratory work. This is a controlled mechanism study, not an LLM-scale benchmark.

HELD-SEED MIXED-SHIFT ESTIMATES

+1.06 pp MLP · +0.92 pp CNN

15/20 MLP pairs and **8/10** CNN pairs favored DoRA. **Holm-adjusted** $p=.382213$ (MLP) / $.158955$ (CNN): internal fixed-checkpoint evidence, not external replication.

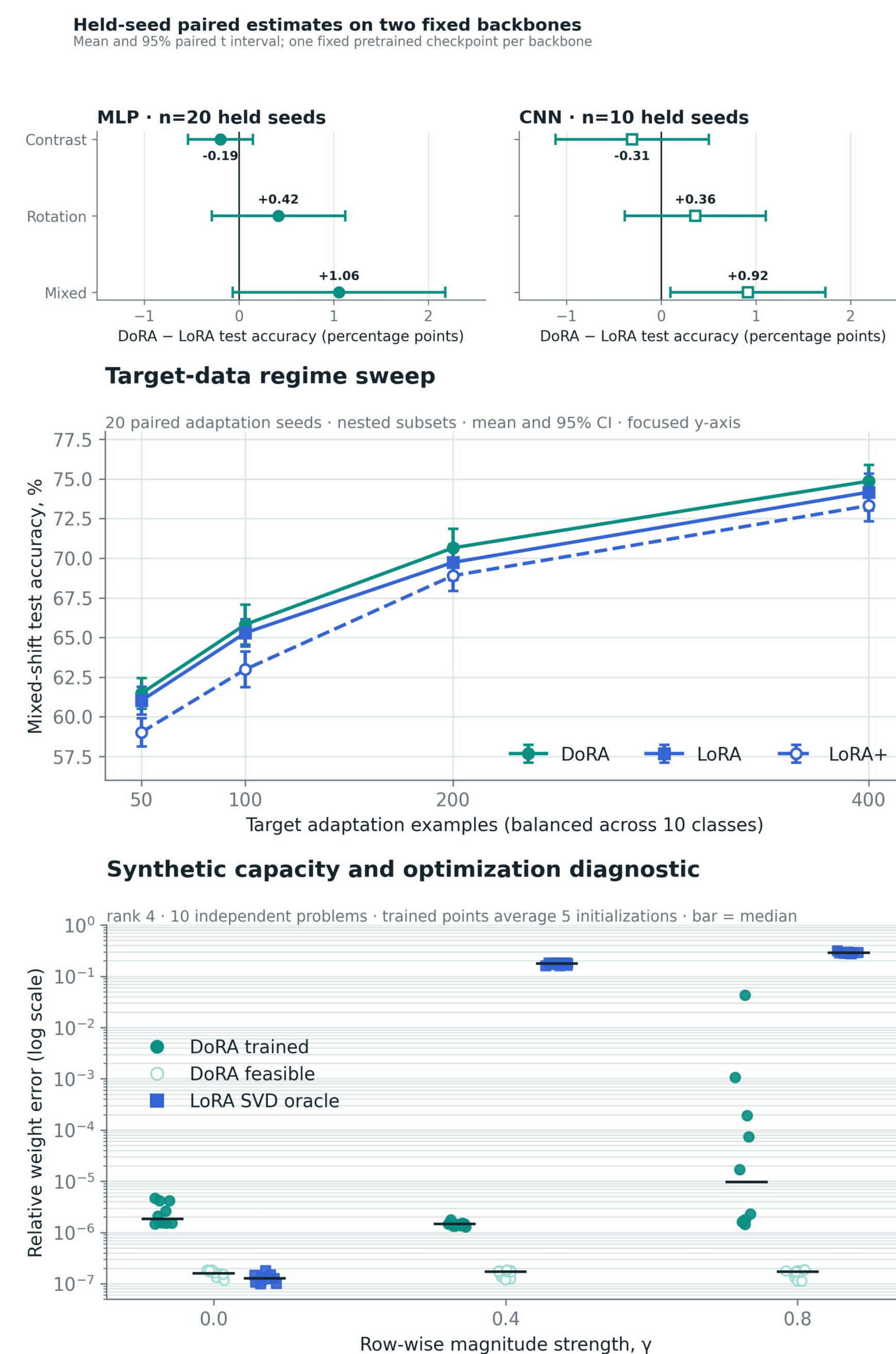


Figure 2. Held-seed estimate → data regime → capacity / optimization diagnostic.

Target-test evaluation followed the frozen validation-selected protocol. Internal confirmation on fixed checkpoints; exact CIs, raw/adjusted p-values, and rank budgets are in the repository.

What the controls still support

- **The mixed-shift DoRA estimate** reaches 75.22% on MLP and 80.53% on CNN - near full FT (75.17%, 80.67%) with only ~12-15% as many trainable parameters.
- **Matched-budget LoRA narrows the MLP gap to +0.53 pp; DoRA–LoRA+ Holm $p=.050935$ ($> .05$); budgeted DoRA +.82 pp (secondary Holm $p=.347089$).**
- **At $\gamma=.8$,** DoRA is representationally exact; training converges below 10^{-3} in 44/50 runs. LoRA's best possible mean error is 0.289.
- **Follow-up hypothesis:** test DoRA when heterogeneous row-norm drift is plausible; retain LoRA and magnitude-only baselines.

[1] Hu et al. LoRA. ICLR 2022. [2] Liu et al. DoRA. ICML 2024.

[3] Hayou et al. LoRA+. ICML 2024. Statistical protocol and raw CSV: GitHub.

Code, validated notebook, full report, raw results, and reproduction instructions → QR.

